Md. Khairul Basar

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Portfolio: https://k-basar.github.io/Portfolios/

EDUCATION

Bachelor of Science in Industrial and Production Engineering, (Feb 2020 – Aug 2025)
Shahjalal University of Science and Technology (SUST), Bangladesh
CGPA: 3.49 / 4.00

RESEARCH PROFILE

Research-oriented Industrial and Production Engineering graduate with a strong foundation in Operations Research, Data Science, Artificial Intelligence, and Machine Learning. Experienced in developing Predict-Then-Optimize frameworks and integrating machine learning with mathematical optimization to solve complex resource allocation, scheduling, and decision-support problems. Passionate about building data-driven, intelligent, and reliable solutions for industrial systems, transportation, and smart manufacturing through optimization, simulation, and AI.

RESEARCH INTEREST

1. **Artificial Intelligence and Machine Learning:** I am interested in developing and applying artificial intelligence and machine learning techniques to solve complex real-world problems in industrial systems, transportation, and decision support. My interests include supervised learning, ensemble methods, deep learning, and data-driven predictive modeling for intelligent automation and operational improvement.
2. **Predict-Then-Optimize Frameworks:** My research focuses on integrating predictive machine learning models with mathematical optimization to create end-to-end decision-support systems. I am particularly interested in uncertainty-aware optimization, where predictive models guide downstream optimization for efficient resource allocation, scheduling, and planning.
3. **Machine Learning-Based Optimization:** I am interested in combining machine learning with optimization techniques to improve planning, scheduling, and resource allocation decisions. My goal is to develop scalable frameworks where machine learning models estimate unknown parameters or objective functions to enhance optimization performance in complex industrial environments.
4. **Data-Driven Decision Support Systems:** I aim to develop intelligent decision-support systems that leverage machine learning, optimization, and data analytics to transform operational data into actionable insights. My interests include predictive analytics, optimization-based planning, and AI-assisted decision-making for industrial and organizational applications.
5. **Simulation and Smart Industrial Systems:** I am interested in simulation-based modeling of manufacturing, transportation, and service systems to evaluate performance under varying operational conditions. My research focuses on integrating discrete-event simulation with machine learning and optimization techniques to identify bottlenecks, improve system efficiency, and support data-driven operational decision-making.

PUBLICATIONS

1. **Optimizing Worker–Task Allocation in a Fish Processing Factory Using a Predict-Then-Optimize Approach**
   - ARPN Journal of Engineering and Applied Sciences (doi: 10.59018/012617)
2. **A Bi-Objective Optimization Model for Real-World Task Allocation in RMG Sewing Lines**
   - 8th IEOM Bangladesh International Conference (**Best Track Paper,** doi: 10.46254/BA08.20250363)
3. **Data-Driven Prediction and Recommendation of Elective Courses for Master's Students**
   - 8th IEOM Bangladesh International Conference (doi: 10.46254/BA08.20250209s)
4. **Maintaining Lane Discipline Can Reduce the Waiting Time in a Road Intersection: A Simulation Study**
   - ICERIE 2025 (doi: https://doi.org/10.2991/978-94-6463-884-4_85)
5. **Determinants of CODP Positioning and Their Effects on Supply Chain Outcomes: An Empirical Investigation in the Furniture Sector**
   - ICERIE 2025 (doi: 10.46254/BA08.20250373)

Projects

1. Excel-ChatGPT Integration: Automating Web Scraping & AI Responses with VBA
   - Integrated ChatGPT with Excel VBA to automate web scraping and instant AI responses. Streamlined data extraction and enhanced workflow automation. Technologies: Excel VBA, OpenAI API, Web Scraping
2. Object Detection Safety System for Industrial Machinery
   - Designed an Object Detection System to enhance workplace safety. Prevents machinery-related hazards by identifying risky movements in real time. Technologies: Computer Vision, Arduino
3. Dual-Prediction System for Traffic Density & Vehicle Energy Consumption Using Machine Learning
   - Built a Python-based ML model for real-time traffic density and fuel consumption prediction. Aids in route optimization and fuel efficiency for urban transportation. Technologies: Python (Pandas, Scikit-learn), Machine Learning, Data Science
4. Pre-Assembly Line Analysssis and Line Balancing at Walton Hi-Tech Industries Ltd.
   - Conducted a detailed Time Study and Motion Analysis on the refrigerator pre-assembly line. Applied line balancing techniques to identify bottlenecks and redistribute workloads effectively. Achieved a significant improvement in line efficiency and reduced idle time. Technologies: Excel, Line Balancing, Work Study, Industrial Engineering
5. Bottleneck Analysis & Scheduling Optimization at Premium Fish & Agro Ind. Ltd.
   - Conducted a bottleneck analysis of the production floor using data-driven methods. Applied scheduling techniques to reduce delays and increase efficiency. Technologies: Excel, Operations Research, Scheduling Algorithms

INDUSTRIAL & JOB EXPERIENCE

Intern, Walton Hi-Tech Industries PLC (May 2025)

• Conducted time study, motion analysis, line balancing, Applied Lean tools

Remote Employee, Industrial 3D Solutions (Dec 2024 – Jun 2025)

• Developed Excel–VBA automation tools, Supported laboratory product supply coordination

Bangladesh Industrial Technical Assistance Centre (BITAC), Dhaka (12 days)

- Gained insights into conventional and non-conventional manufacturing processes.

Khadim Ceramics, Sylhet (1 day)

- Acquired exposure to industrial ceramic manufacturing processes.

TECHNICAL SKILLS

- **Programming & Data Science:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, LangChain, CrewAI
- **Machine Learning:** Statistical ML Methods, Deep Learning, NLP, Computer Vision, Graph Neural Networks (GNNs), Agentic Decision-Making, Agentic LLMs, AI Reasoning, RAG
- **Optimization & Simulation:** LP, NLP, MILP, Convex Optimization, Discrete Optimization, Simulation Modeling, Gurobi, Scheduling, Arena Simulation

CERTIFICATIONS

- Introduction to Data Science with Python – EDGE Bangladesh
- Introduction to Deep Learning with PyTorch – DataCamp
- Excel/VBA for Creative Problem Solving – University of Colorado Boulder
- Six Sigma White Belt – CSSC

Achievements

- Awarded a 60% scholarship for the **ISCEA Prize Case Competition 2024**
- Finalist in **IPE Case Quest 1.0**

Interests

Poetry, Chess, Reading, Travelling

References

- **Dr. Abul Mukid Mohammad Mukaddes**
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